



Comparative biochemical and histopathological evaluations proved that receptacle is the most effective part of *Cynara scolymus* against liver and kidney damages



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ABSTRACT

Ethnopharmacological relevance: The liver and kidney are among the most important organs in the body, where metabolic and elimination functions take place. During this process, liver and kidneys may suffer damage due to ingestion or formation of toxic metabolites leading to organ loss and even death. Artichoke (*Cynara scolymus* L.) leaf has long been recognized as a popular herbal remedy in traditional medicines with beneficial effects on liver. **Aim of the study:** In phytotherapy leaves are the part used to support the liver functions and for treatment of damage induced by various toxins, while fleshy receptacle is cooked as meal to support liver homeostasis. However, effects of other plant parts on liver such as stems, bracts have not much attracted the attention of scientific community so far. In this study we investigated comparatively the hepatoprotective and nephroprotective effects of different plant parts of artichoke, i.e. receptacles, outer bracts, inner bracts, and stems with that of leaves upon paracetamol-induction in rats. **Materials and methods:** Aqueous ethanol (80%) extracts obtained from the different parts of artichoke were administered for five consecutive days after paracetamol induction to rats. At the end of experimental period blood samples from the experimental animals were taken for biochemical tests, while livers and kidneys were removed for further histopathological evaluation. **Results:** The histopathological examinations of liver and kidney tissues revealed that the receptacle and stem extracts of the artichoke were the most effective parts by improving the experimentally induced pathology in both liver and kidney. Biochemical tests also supported the histopathological data; receptacle, stem and bract extracts reduced serum alanine transaminase (ALT) and aspartate transaminase (AST) levels, but not alkaline phosphatase (ALP), creatinine and blood urea nitrogen (BUN) levels. **Conclusions:** Histopathological and biochemical studies have shown that receptacle and stem extracts of artichoke were found to exert higher protective activity on liver and kidney damage induced by paracetamol comparing to its bract and leaf extracts, the latest is officially recognized as herbal remedy.

1. Introduction

The liver is one of the most crucial organs in the human body and plays role in maintaining and regulation of the homeostasis. Failure of liver, termed as hepatotoxicity, is mostly caused by chemical agents such as environmental toxins (Zimmerman and Lewis, 1995) or pharmacotherapy (Bjornsson, 2016). Particularly drug-induced

hepatotoxicity is reported as a serious health problem leading to withdrawal of approved medications from the market (Pugh et al., 2009).

Several herbal extracts were suggested to have protective effects against drug-induced hepatotoxicity. Among these Silymarin, derived from the seeds of *Silybum marianum*, has been reported to protect hepatocytes from various toxins such as paracetamol or carbon

Abbreviations: ALT, alanine transaminase; ALP, alkaline phosphatase; AST, aspartate transaminase; BUN, blood urea nitrogen; CMC, carboxymethyl cellulose

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tetrachloride (CCl₄) and healing benefits were also proven clinically (Wah Kheong et al., 2017). Artichoke (*Cynara scolymus*) leaf extract is another popular herbal remedy to support liver functions in traditional folk medicine, and its hepatoprotective and antioxidative effects have been previously proven both experimentally and clinically (Adzet et al., 1987; Gebhardt, 1997; Gebhardt and Fausel, 1997; Speroni et al., 2003; Ben Salem et al., 2015; Colak et al., 2016; Rangboo et al., 2016). Pre-treatment of rats with artichoke leaf extract for two weeks via oral gavage significantly reduced the plasma transaminase enzyme activity induced upon CCl₄ treatment (Mehmetçik et al., 2008). Phenolics (Sanchez-Rabaneda et al., 2003; Schutz et al., 2004) and terpenes (Fritsche et al., 2002; Shimoda et al., 2003) are reported to be the main active ingredients. In spite of popularity and widespread use of artichoke leaves as a hepatoprotective remedy in liver complaints, effect of other parts of the plant, i.e., stem, bracts, or receptacle has not been studied comparatively.

On the other hand, in spite of a high number of reports on the liver toxicity induced by acute administration of paracetamol (Mandal et al., 2000; Larson et al., 2005; Wang et al., 2010; Ahmad et al., 2019), its toxic effects on kidneys are less known (McGill and Jaeschke, 2019; Canayakin et al., 2016; Mudge et al., 1978). Nephrotoxicity episodes due to paracetamol administration has been mainly attributed to cytochrome P-450 mixed function oxidase isoenzymes present in the kidney, while several other mechanisms have also been reported, including the role of prostaglandin synthetase and N-deacetylase enzymes (Mazer and Perrone, 2008). However only few studies have evaluated the protective effects of natural medicines against both the liver and the kidney toxicity of paracetamol simultaneously (Menyiy et al., 2018).

This study, for the first time, focused on the comparatively efficacy of artichoke extracts obtained from its different parts including receptacle, inner bracts, outer bracts, leaves, and stem on paracetamol-induced acute hepatotoxicity and nephrotoxicity.

2. Materials and Methods

2.1. Chemicals

Carboxymethyl cellulose (CMC), ethanol, paracetamol and toluene were purchased from Sigma Aldrich. Hematoxylin-Eosin were purchased from DDK (Italy). Ketamine HCl was purchased from Richter Pharma AG and *Silybum marianum* extract (Legalon®, Madaus).

2.2. Plant material and preparation of plant extracts

2.2.1. Plant material

The *Cynara scolymus* L. plant materials were harvested on May 12, 2016 in İzmir, Turkey from the cultivar area (YUEF, 2016.008). Plant materials were identified by Dr. E. Celep from Yeditepe University. Harvested whole plant was dried at room temperature for two weeks under shed and separated into its botanical parts (receptacle, inner bracts, outer bracts, leaves, and stem). Dried materials were stored at deep freezer and ground to powder in a mechanic grinder and sieved before extraction.

2.2.2. Extraction procedure

Extraction process of each plant material was carried out with Ethanol:water (8:2) mixture by continuous mixing in a reciprocating shaker for 2 h. Then the residual marcs were separated by filtration and each of the filtrated extract was condensed to dryness in a rotary evaporator to remove ethanol. Each residue was reconstituted with water, kept at -80 °C overnight, and then lyophilized. The extract yields were given in Table 1.

2.2.3. Preparation of treatments

The CMC solution (0.5%) was used as a negative control. *Silybum marianum* L. extract (100 mg/kg) as the reference drug and the

Table 1

Extraction yields of various parts of *Cynara scolymus*.

Part of Plant	Weight of Powdered Plant (g)	Weight of Lyophilized Extract (g)	Yield %
Receptacle	55	5.72	10.40%
Outer Bract	178	9.26	5.20%
Inner Bract	52	4.66	8.96%
Stem	28	3.61	12.89%
Leaves	55	9.93	18.05%

lyophilized artichoke extracts (500 mg/kg) were dispersed in 0.5% CMC prior to testing.

2.3. Animals

Male Sprague-Dawley rats were obtained from the Yeditepe University Medical School Experimental Research Center (YUDETAM) and housed at controlled room temperature (21 ± 1 °C) with a 12:12 h light-dark cycle in polypropylene cages with bedding. The experimental protocol was approved by the Ethic Committee of Yeditepe University. Rats (10 weeks old) were randomly divided into 7 groups with 7 animals in each group. The groups and corresponding treatments are indicated Table 2. Paracetamol as the hepato- and nephrotoxin is dispersed in 0.5% CMC mixture and administered orally (2000 mg/kg) (Mishra et al., 2015; Roy et al., 2015; Haidara et al., 2019).

2.4. Biochemical analyses

At first day, just after collecting the blood samples from each animal to determine the basal biochemical levels, 2000 mg/kg paracetamol suspension was administered orally to induce renal- and hepato-toxicity (Roy et al., 2015; Ozatik et al., 2019). Afterwards, 2 mL of each artichoke extract in 0.5% CMC was orally administered to animals in Groups Res, OB, IB, St, and Lf (500 mg/kg dose). While 0.5% CMC and *Silybum marianum* extract in 0.5% CMC were administered to Cont. and SM groups, respectively. Administration of the extracts were carried on for three consecutive days. 0.8 mL blood sample of each rat was drawn per day via vena jugularis externa. On the fifth day (Canayakin et al., 2016), animals were sacrificed with a high dose of anesthesia (with ketamine HCl; 100 mg/kg). Then liver and kidney tissues were collected for further analyses.

Blood samples were drawn to analyze the biochemical parameters, i.e., the levels of ALT, AST, blood urea nitrogen (BUN), creatinine, alkaline phosphatase (ALP). On the first day, blood samples were taken before the hepatotoxicity induction to determine the basal levels. Then, until sacrifice of the experimental animals, blood samples were taken periodically on every 24 h.

2.5. Histopathological examinations

Liver and kidney samples were collected in 10% formaldehyde solution and stored at 4 °C for histopathological analyses. Tissue samples

Table 2

The groups of animals and applications.

Group	Application
Res	Receptacle extract of <i>Cynara scolymus</i> via oral gavage at 500 mg/kg dose
OB	Stem extract of <i>Cynara scolymus</i> via oral gavage at 500 mg/kg dose
IB	Inner Bract extract of <i>Cynara scolymus</i> via oral gavage at 500 mg/kg dose
St	Outer Bract extract of <i>Cynara scolymus</i> via oral gavage at 500 mg/kg dose
Lf	Leaves extract of <i>Cynara scolymus</i> via oral gavage at 500 mg/kg dose
Cont.	2 mL 0.5% CMC was applied for negative control
SM	<i>Silybum marianum</i> extract in 0.5% CMC was applied at 100 mg/kg dose for positive control

have first been washed with tap water for 2 h and dehydrated by treating with 70%, 90%, 96% and 100% ethyl alcohol, successively, after fixation. Samples then were gotten transparent with toluene and embedded in paraffin after overnight incubation at 60 °C in the oven. Five slices with 5 µm thickness were obtained and each slice was stained with Hematoxylin-Eosin (H&E).

2.5.1. Histopathological scoring of kidney samples

Each slice was assessed in terms of Bowman cavity, glomerular degeneration, proximal and distal tube degenerations, vascular congestion and interstitial edema. Each assessment for every animal was scored between 0-3 (0: no change, 1: mild, 2: moderate, 3: severe). Maximum score for each sample was 12 whereas minimum score was 0. For each organ in each animal, the scores from different parameters were added up, and mean histopathological score of the organ was determined.

2.5.2. Histopathologic scoring of liver samples

Liver samples were prepared to facilitate comparisons of the same liver lobes of all animals. Each slice was assessed in terms of hepatic vascularization, enlargement in pyknotic nuclei and sinusoids, Kupffer cell infiltration and vascular congestion. Each assessment was scored between 0-3 (0: no change, 1: mild, 2: moderate, 3: severe). Maximum score for each sample was 12 whereas minimum score was 0.

2.6. Statistical analyses

The results were expressed as mean $\pm$ standard deviation (SD) in the figures and mean $\pm$ standard error of the mean (SEM) in the tables. For the analyses of the histological examinations one-way ANOVA followed by Tukey Multiple Comparison test and for the biochemical analyses two-way ANOVA followed by Tukey Multiple Comparison test were applied. The differences between groups were compared using a Two-Way ANOVA followed by Tukey Multiple Comparison test either within days, comparing the treatments or within treatment groups, and comparing the days. GraphPad Prism® 6 was used to perform statistical analysis. $p < 0.05$ was considered statistically significantly different.

3. Results

3.1. Extraction yields

Extraction yields were given in Table 2.

3.2. Histopathologic examinations

Number of survived animals, scores that are the indicators for pathological changes in kidney including degeneration of Bowman space and glomeruli, degeneration of proximal and distal tubules, vascular congestion, interstitial edema and for the liver including vacuolization of hepatocytes and pyknotic nuclei, enlargement of sinusoids,

enlargement sinusoids, Kupffer cell infiltration, vascular congestion were indicated in Table 3.

3.2.1. Histopathological liver examinations

Histopathological liver scores of groups Res. and St were statistically lower than other groups ($p < 0.001$). No other statistically significant differences were found between the groups (Fig. 1).

3.2.2. Histopathological kidney examinations

Histopathological kidney scores of Group Res. were significantly lower than the other groups except the Group SM ($p < 0.001$). Also, lower histopathologic kidney scores were detected in groups OB, IB, St., but not in group Lf when compared to group Cont. as a negative control group. Furthermore, histopathological kidney score of group SM as a positive control group (*Silybum marianum* extract treated group) were found to be lower than group Cont. (Fig. 2).

3.3. Biochemical analyses

In order to evaluate the effect of extracts on the liver and kidney, clinically important biochemical serum parameters i.e., AST, ALT, ALP, BUN and creatinine levels were analyzed. As shown in Table 4, paracetamol administration increased the ALT levels in all groups. This increase was alleviated in all groups five days following the administration of paracetamol. On the other hand, at day 2, ALT levels were significantly higher in SM group when compared to other groups except group Cont. In addition, group Cont. exhibited significantly higher ALT levels compared to group St and IB. The measurements revealed that ALT levels on days 3–5 were not significantly different in any groups, except for the group SM that exhibited significantly higher ALT compared to other groups. When the AST levels were examined, the increased AST levels by the paracetamol decreased after five days of administration in all groups, but this decrease was not significant compared to day two except for OB extract ($p < 0.05$, Table 5). At day 2, AST levels of OB group were found to be elevated significantly compared to St and IB groups, while no significant differences were observed between the groups on day 3–5.

On the other hand, all artichoke extracts except Lf were found to be ineffective to maintain the serum ALP levels (Table 6). On the other hand, ALP levels before the toxicity induction were significantly higher in the group Lf compared to group St and Cont. At day 3 and 4, group Lf exhibited significantly lower ALP levels compared to group Cont., while no statistically significant inhibitory rate provided by leaf extract were detected compared to other groups.

As for the serum kidney markers; increased levels of BUN and creatinine after paracetamol induction were improved by all the artichoke extracts except receptacle after 5 days of treatment (Table 7-a and -b). However, these rates were not statistically significant.

Table 3
Histopathological evaluations of experiment groups.

		Res	OB	IB	St	Lf	Cont.	SM
Number of living animals at the end of the experiment (n)		5	7	6	7	4	6	7
Kidney	Degeneration of Bowman Space and Glomeruli	0/5	5/7	6/6	2/7	3/4	6/6	2/7
	Degeneration of Proximal and Distal Tubules	0/5	4/7	6/6	5/7	4/4	6/6	3/7
	Vascular Congestion	0/5	5/7	4/6	4/7	4/4	6/6	2/7
	Interstitial Edema	0/5	5/7	3/6	4/7	4/4	6/6	3/7
Liver	Vacuolization of Hepatocytes and Pyknotic Nuclei	2/5	2/7	5/6	6/7	4/4	5/6	5/7
	Enlargement Sinusoids	0/5	0/7	5/6	6/7	3/4	5/6	5/7
	Kupffer Cell Infiltration	0/5	0/7	5/6	6/7	4/4	6/6	5/7
	Vascular Congestion	0/5	0/7	6/6	7/7	4/4	6/6	4/7

Abbreviations: Res: receptacles; OB: Outer bracts; IB: Inner bracts; St.: Stems; Lf: Leaves; Cont: Negative control; SM: *Silybum marianum* extract (positive control).

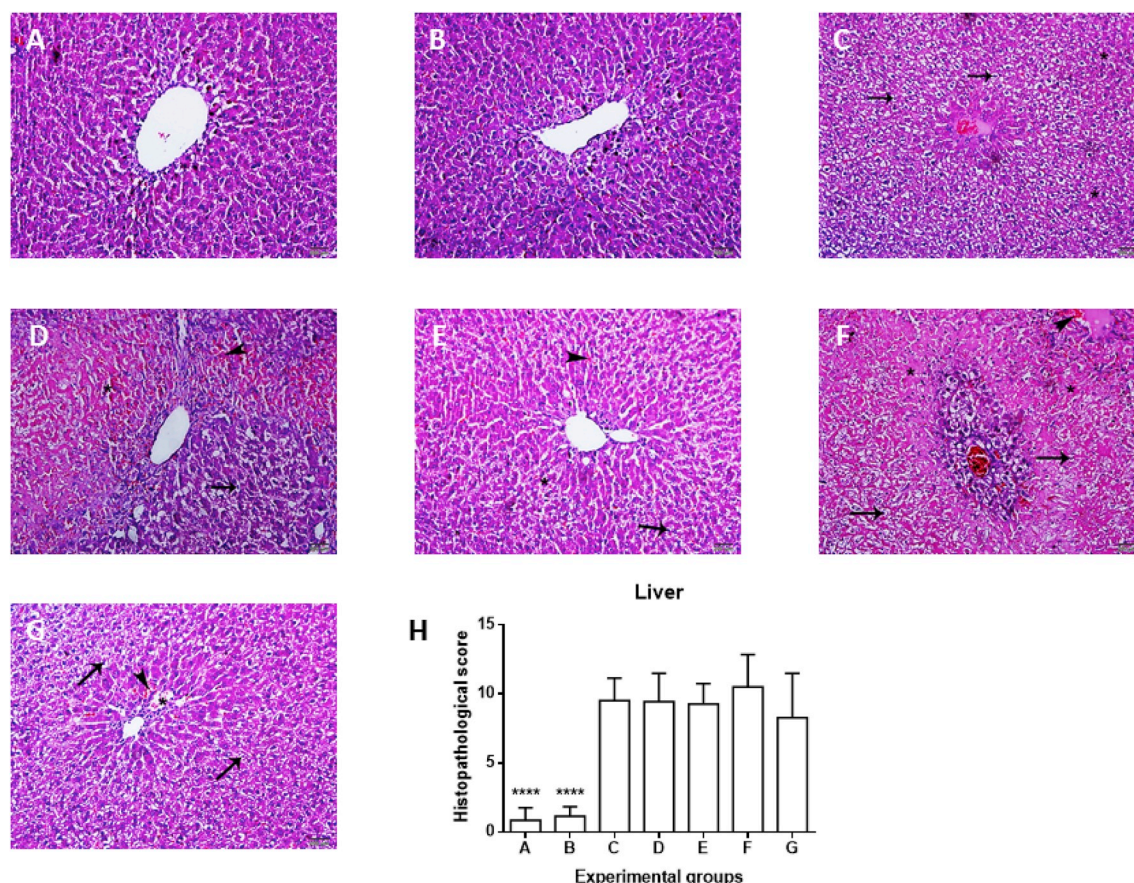


Fig. 1. Histopathological examination of the liver tissue.

A. Hepatocytes and sinusoids are seen in radically located hepatic lobules in a uniform morphology (Group Res.). **B.** Hepatic lobule in the normal structure of the liver and hepatocytes and sinusoids located radially in the liver are seen (Group St.). **C.** Hepatic lobule of the liver with impaired liver and picnic nucleus hepatocytes (arrow) and sinusoidal structures (*) are seen to be impaired (Group IB). **D.** Hepatic lobules of the liver are seen as hemorrhagic (arrow head) areas in which the hepatic lobule is impaired (OBSt.). **E.** Hepatic lobule liver of the liver and picnic nucleus hepatocytes (arrow), sinusoidal structures (*) are damaged, and hemorrhagic (arrow head) areas are seen (Group Lf). **F.** In the liver, there are hepatic lobules that are damaged extensively and hemorrhagic and picnic nucleus hepatocytes (arrow) and impaired sinusoidal structures (*) (arrow head) are observed (Cont. Group). **G.** In the liver, picnic nucleus hepatocytes (arrow) and sinusoidal structures (*) are impaired and scattered hemorrhagic (arrow head) areas are seen (Group SM). **H.** Mean histopathological scores. Data are mean $\pm$ S.D. Statistical analysis: One-way ANOVA followed by the Tukey's multiple comparison test (****p < 0.0001).

4. Discussion

In this study, potential protective effects of artichoke extracts prepared from different parts of the plant on the paracetamol-induced hepatotoxicity and nephrotoxicity were investigated, comparatively. Paracetamol is a widely recognized and commonly used drug with analgesic and antipyretic activities (Graham et al., 2013). Hepatotoxicity and nephrotoxicity caused by the administration of paracetamol are widely reported and generally limited to mild lesions in case of acute use at therapeutic doses, however, use in higher doses may lead to serious damage of liver and kidneys (Toussaint et al., 2010).

The metabolism and elimination of paracetamol take place in the liver and cytochrome P450 family of enzymes catalyze its conversion to N-acetyl-p-benzoquinone imine (NAPQI) (Dahlin et al., 1984). NAPQI is the reactive metabolite of paracetamol and detoxified when conjugated with glutathione (McGill et al., 2010). Presence of excess amount of NAPQI causes the block of the metabolic pathways leading to mitochondrial dysfunction and oxidative stress (McGill et al., 2012). The most common unfavorable impact of the paracetamol is the hepatic damage, however, its damage to the kidney was also noticed (Mudge et al., 1978). Metabolism of paracetamol to p-aminophenol by deacylation in kidneys leading to nephrotoxicity (Newton et al., 1986). Recent studies indicated the nephrotoxicity may be caused by free and non-protein derivatives of paracetamol-cysteine complexes (Stern et al.,

2005a and 2005b). Breakdown of paracetamol glutathione conjugates to these metabolites by γ -glutamyltransferase, an enzyme abundantly found in kidney, leads to depletion of the renal glutathione by free paracetamol-cysteine complexes.

As the most efficient treatment for paracetamol overdose is to replenish the liver glutathione stocks, N-acetyl cysteine (NAC) administration to the patients is practiced in the clinics. The elevated levels of glutathione lead to scavenging the excess NAPQ1 and interference with protein binding. In addition to the NAC administration, charcoal or emetic agents are advised to be helpful in management of paracetamol overdose (Spiller et al., 2006, Spiller and Sawyer, 2007).

Herbal drugs for the treatment of liver disorders have revealed promising results (Asadi-Samani et al., 2015). Among these, artichoke (*Cynara scolymus*) leaf is one of the most popular plants used in the treatment of liver and bile disorders. It was reported that administration of the artichoke leaf extract to rats significantly reduced the transaminase activity and alleviated the histopathological changes (Mehmetcik et al., 2008). The effect of its extract could be attributed to the rich antioxidant ingredients such as mono- and dicaffeoylquinic acids (cynarin and chlorogenic acid), caffeic acid and flavonoids including luteolin glycosides (-7- β -rutinoside; -7- β -glucoside and -4- β -glucoside). Beneficial effects of artichoke in the control of fatty liver and hyperlipidemia in rats and in humans were also evidenced (Bundy et al., 2008). Previous studies have revealed that LDL oxidation was

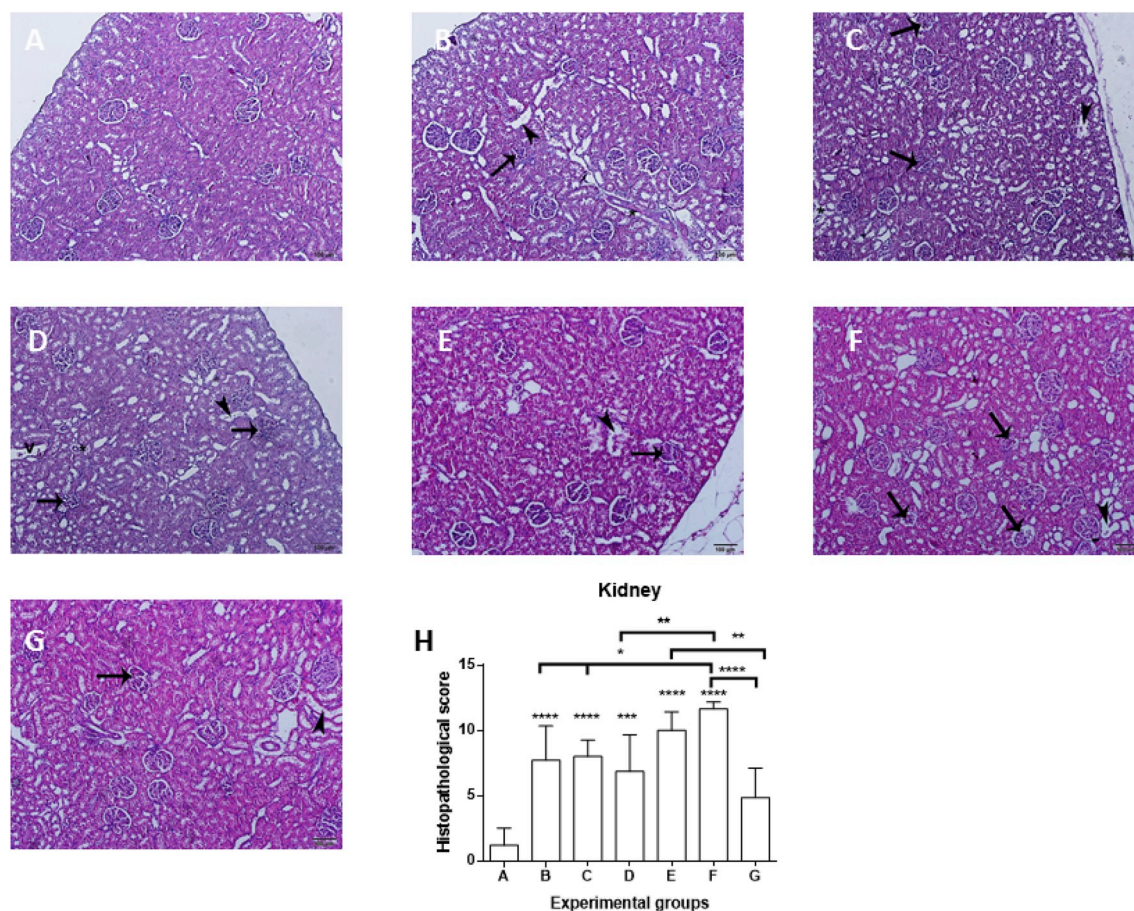


Fig. 2. Histopathological findings for kidney tissue.

A. Regular morphologic glomeruli and tubules are seen in group Res. in the kidney cortex stroma. **B.** Some glomeruli (arrows) and tubules (arrowheads) are damaged, edema (*) in Group St in the kidney cortex. **C.** A large number of glomeruli (arrows) in group IB in the kidney cortex, and tubules (Arrow head) damage, edema (*) is seen. **D.** There are numerous glomeruli (arrows) and tubules (arrowheads) damage, edema (*), vasoconstriction (>) in the OB group in the kidney cortex. **E.** Large numbers of glomeruli (arrows) and tubules (arrowheads) are seen in group Lf. in the kidney cortex. **F.** There are numerous glomeruli in the Cont. group of the kidney cortex and damage to the Bowman capsule (arrow) and tubules (arrowhead). **G.** There are few glomeruli (arrows) and tubules (arrowheads) in group SM in the kidney cortex. **H.** Mean histopathological scores. Data are mean $\pm$ S.D. Statistical analysis: One-way ANOVA followed by the Tukey's multiple comparison test. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ and **** $p < 0.0001$.

Table 4

Effect of the extracts from different parts of artichoke on ALT levels (U/L).

Days	Res.(n = 5)	St (n = 7)	IB (n = 6)	OB (n = 7)	Lf (n = 3)	Cont. (n = 6)	SM (n = 6)
1	48.00 $\pm$ 2.17	53.71 $\pm$ 2.8	56.33 $\pm$ 3.83	51.71 $\pm$ 2.36	55.33 $\pm$ 7.96	48.50 $\pm$ 2.64	42.33 $\pm$ 2.70
2	1357 $\pm$ 1110	544.30 $\pm$ 219.80	653.8 $\pm$ 118.70	1722 $\pm$ 828.40	871.00 $\pm$ 261.30	2454 $\pm$ 1293 ^{b, c}	3470 $\pm$ 1013 ^{a, b, c, d, e}
3	1051 $\pm$ 338.80	1205 $\pm$ 449.50	884.8 $\pm$ 302.70	1124 $\pm$ 343.70	1444 $\pm$ 244.60	1494 $\pm$ 703.50	1966 $\pm$ 122.10
4	163.20 $\pm$ 43.68	245.40 $\pm$ 87.95	175.70 $\pm$ 59.45	239.40 $\pm$ 67.96	244.70 $\pm$ 1.20	270.20 $\pm$ 112.9	301.20 $\pm$ 35.49
5	82.80 $\pm$ 13.22	90.14 $\pm$ 24.73	86.00 $\pm$ 17.63	82.86 $\pm$ 16.42	88.00 $\pm$ 21.13	99.67 $\pm$ 43.75	98.00 $\pm$ 11.94

Table 5

Effect of the extracts from different parts of artichoke on serum AST levels (U/L).

Days	Res. (n = 5)	St (n = 7)	IB (n = 6)	OB (n = 7)	Lf (n = 3)	Cont. (n = 6)	SM (n = 6)
1	84.00 $\pm$ 3.91	74.29 $\pm$ 3.41	73.83 $\pm$ 3.89	63.43 $\pm$ 1.69	75.33 $\pm$ 2.85	71.50 $\pm$ 5.66	62.67 $\pm$ 1.88
2	1889 $\pm$ 1130.00	1464 $\pm$ 613.30	1514 $\pm$ 265.90	4237 $\pm$ 1954.00 ^{b, c}	2740 $\pm$ 1438.00	2172 $\pm$ 492.60	3695 $\pm$ 1164.00
3	1589 $\pm$ 503.30	1888 $\pm$ 494.40	1461 $\pm$ 492.00	2000 $\pm$ 592.50 ^x	2438 $\pm$ 496.70	1848 $\pm$ 794.40	3936 $\pm$ 604.40
4	147.60 $\pm$ 35.43	282.60 $\pm$ 88.42	220.70 $\pm$ 61.30	272.10 $\pm$ 72.66 ^y	276.70 $\pm$ 3.38	275.20 $\pm$ 88.42	348.80 $\pm$ 60.87
5	105.60 $\pm$ 14.83	104.70 $\pm$ 11.62	117.70 $\pm$ 6.55	113.00 $\pm$ 15.71 ^z	156.00 $\pm$ 54.05	152.80 $\pm$ 54.73	128.80 $\pm$ 16.49

Data are mean $\pm$ SEM. Statistical analysis: Two-way ANOVA followed by Tukey Multiple Comparison test. ^{a, b, c, d, e, f, g, x, y} and ^z $p < 0.05$ as ^ap vs. Res; ^bp vs. OB; ^cp vs. IB; ^dp vs. St; ^ep vs. Lf; ^fp vs. Cont.; ^gp vs. SM extracts and ^{x, y} and ^zp vs. day 2.).

Table 6

Effect of the extracts from different parts of artichoke on serum ALP levels (units/L).

Days	Res. (n = 5)	St (n = 7)	IB (n = 6)	OB (n = 7)	Lf (n = 3)	Cont. (n = 6)	SM (n = 6)
1	381.20 ± 37.58	295.90 ± 26.82	389.20 ± 45.28	368.00 ± 11.09	458.30 ± 17.23 ^{b, f}	285.30 ± 52.77	366.00 ± 44.39
2	258.60 ± 22.35	259.90 ± 15.90	277.00 ± 35.07	300.30 ± 10.36	283.00 ± 41.51	274.50 ± 28.44	286.5 ± 34.40
3	290.00 ± 30.45	354.40 ± 16.25	313.70 ± 42.96	366.40 ± 20.41	235.00 ± 36/53 ^g	346.2 ± 54.11	410.30 ± 27.00
4	242.60 ± 33.02	241.00 ± 20.04	221.80 ± 34.55	237.70 ± 16.69	152.70 ± 20.34 ^g	233.7 ± 26.28	312.8 ± 34.11
5	225.20 ± 36.88	172.70 ± 19.95	165.70 ± 24.64	159.40 ± 20.02	124.00 ± 19.14	131.20 ± 24.33	237.70 ± 26.55

Data are mean ± SEM. Statistical analysis: Two-way ANOVA followed by Tukey Multiple Comparison test (^{a, b, c, d, e, f, and g} $p < 0.05$ as ^ap vs. Res; ^bp vs. OB; ^cp vs. IB; ^dp vs. St; ^ep vs. Lf; ^fp vs. Cont.; ^gp vs. SM extracts.).

Table 7

Effect of the extracts from different parts of artichoke on the serum kidney markers; BUN and creatinine levels (mg/dL).

Days	Res.(n = 5)	St (n = 7)	IB (n = 6)	OB (n = 7)	Lf (n = 3)	Cont. (n = 6)	SM (n = 6)
a. Effect on BUN levels							
1	22.20 ± 0.49	20.29 ± 0.71	18.67 ± 0.67	18.29 ± 0.36	17.33 ± 0.33	17.83 ± 0.31	17.17 ± 0.60
2	15.40 ± 0.68	14.86 ± 0.34	23.83 ± 8.25 ^{a, b, d, g}	15.43 ± 0.48	15.33 ± 2.33	17.67 ± 0.95	15.67 ± 0.49
3	12.80 ± 1.24	13.00 ± 0.58	12.83 ± 0.79	11.00 ± 0.58	11.67 ± 1.20	11.17 ± 0.70	12.83 ± 0.83
4	15.00 ± 0.83	13.57 ± 1.11	14.50 ± 1.26	12.00 ± 1.59	9.33 ± 2.03	12.83 ± 1.08	11.50 ± 0.50
5	22.20 ± 3.34	16.71 ± 1.34	14.50 ± 0.81	18.00 ± 0.79	12.00 ± 1.52 ^a	15.67 ± 0.42	17.50 ± 1.38
b. Effects on creatinine levels							
1	0.468 ± 0.014	0.478 ± 0.012	0.455 ± 0.016	0.434 ± 0.009	0.423 ± 0.003	0.446 ± 0.004	0.441 ± 0.006
2	0.494 ± 0.021	0.458 ± 0.0299	0.495 ± 0.011	0.484 ± 0.007	0.480 ± 0.015	0.528 ± 0.009 ^b	0.506 ± 0.012
3	0.470 ± 0.013	0.491 ± 0.008	0.490 ± 0.024	0.462 ± 0.004	0.430 ± 0.011	0.466 ± 0.013	0.476 ± 0.008
4	0.456 ± 0.009	0.494 ± 0.006	0.483 ± 0.017	0.465 ± 0.010	0.400 ± 0.011 ^{b, c, f}	0.473 ± 0.013	0.466 ± 0.004
5	0.488 ± 0.027	0.467 ± 0.017	0.465 ± 0.011	0.470 ± 0.017	0.453 ± 0.022	0.433 ± 0.013	0.480 ± 0.015

Data are mean ± SEM. Statistical analysis: Two-way ANOVA followed Tukey Multiple Comparison test (^{a, b, c, d, e, f, and g} $p < 0.05$ as ^ap vs. Res; ^bp vs. OB; ^cp vs. IB; ^dp vs. St; ^ep vs. Lf; ^fp vs. Cont.; ^gp vs. SM extracts.).

decreased by administration of artichoke leaf extract (Zapolska-Downar et al., 2002) and beneficial effects on the serum cholesterol, triglyceride levels and in the hepatic and cardiac pro-oxidant status were also reported (Kucukgergin et al., 2010).

Increased levels of AST and ALT in the serum are the indicators of hepatic damage (Aubrecht et al., 2013). In addition, elevated circulation of alkaline phosphatase (ALP) was observed in the serum connected to hepatobiliary injury (Aubrecht et al., 2013). On the other hand, serum creatinine and blood urea nitrogen (BUN) levels are particularly used for identification of the kidney damage (Rahman et al., 2018).

In a previous study, the possible protective effect of artichoke on liver and kidney functions upon Cadmium (Cd)-induced toxicity was investigated and it was reported that its leaf extract to exert both nephroprotective and hepatoprotective properties (El-Boshy et al., 2017). The AST, ALT, creatinine and BUN levels were significantly reduced by the leaf extract when compared to that of the Cd-treated group, supporting the effect of artichoke leaf on liver and kidney protection (El-Boshy et al., 2017). In another study, potential of artichoke leaf against gentamycin-induced kidney damage was also reported (Khatab et al., 2016).

Although leaf is the part used in healing practice for liver diseases since long, effects of other parts of artichoke only recently attracted the attention of scientific community. In a previous preliminary study, we found that the 80% ethanol extract from the receptacle, stem and bracts of artichoke reduced the serum ALT and AST levels in carbon tetrachloride (CCl₄)-induced hepatotoxicity model in mice. Inhibitory percentages were found to be 66, 64 and 66% for ALT and 32, 44 and 34% for AST, respectively (Aktay et al., 2000). Recently, El Sayed et al., [2018] also investigated the antihepatotoxic effects of artichoke extracts from receptacles, bracts and roots comparatively with that of leaves. They also used the same doses (500 mg/kg) as we did in the present study, but hepatotoxicity was induced by CCl₄ administration twice weekly for one month. They observed highest inhibitory rates with receptacle extract on serum ALT (24%), AST (18%) and ALP (19%) levels, then followed by bracts, leaves and roots, successively.

This study, aimed to investigate the hepatoprotective and nephroprotective effects of the extracts obtained from individual artichoke parts, i.e., receptacles, stems, outer bracts and inner soft bracts comparatively with those of leaves. Among these receptacles are used as food, while stems and inner bracts are also edible parts while fresh, but outer bracts are inedible due to its hard fabric. In the present study, we observed reduction in the serum ALT and AST levels of all artichoke extracts after the paracetamol challenging, but not in ALP, creatinine and BUN levels. After the 5 days of treatment the order of inhibitory rates in the serum ALT levels were for receptacle (17%), stem (16.9%), inner bracts (13.7%), leaves (11.7%), and outer bracts (1.0%) in the order of potency, but none of these values was significantly different than the control group. For the same administration period inhibitory rates for AST levels were as follows; for outer bracts (31.5%), receptacle (30.9%), stem (26.0%), and inner bracts (23.0%), but no inhibition was observed with leaves. The reason that we did not see any difference between the results might be due to the duration of the treatment. In the study conducted by El Morsy and Kamel (2015) rats were treated with artichoke leaf extract for 14 days upon paracetamol induction and they observed significant reductions in ALT and AST level in the extract treated animals. Another study also reported the reduction in the AST, ALT, creatinine and urea levels upon 4 weeks of artichoke leaf extract administration (Kucukgergin et al., 2010). Therefore, longer treatments with the extracts may lead to significant improvements in the blood parameters.

Although serum levels of biochemical markers for liver and kidney were not significant enough to prove a sufficient activity for the extracts from different parts of artichoke, contrarily histopathological data proved that particularly the receptacle and stem extracts of *Cynara scolymus* alleviated significantly the pathological changes induced by paracetamol in the liver and kidneys. In parallel with liver, kidney histopathology was also significantly improved by the artichoke extracts and receptacle was the most effective with this regard.

As conclusion, although the basal leaves of artichoke have long been officially recognized as an herbal remedy against hepatic disorders in phytotherapy, effects of other parts of the plant on liver functions have

not received much attention of the scientific community. Firstly Aktaş et al. (2000) reported the strong inhibitory effects of receptacle, bracts and stems, but no comparison was made with the leaf extract. After this report only recently, El Sayed et al. (2018) comparatively investigated the effects of receptacle, roots and bracts with that of leaves and found higher effects of receptacle and bracts than leaves. These two studies confirm the results of present study that receptacle provides the highest antihepatotoxic activity among the all investigated parts of artichoke. El Sayed et al. (2018) also reported that polyphenolic content of receptacles was found two times higher than bracts and four times than leaves and antioxidant activity potency of these extracts were also in the same order. Accordingly, it is evident that artichoke receptacle exhibits potent activities against liver and kidney damages induced by hepatotoxins and nephrotoxins, in fact these activities were much higher than the popular liver remedy *Silybum marianum* seeds. On the other hand, stem was also found more efficient than leaves on hepatic damages.

The results of this experimental study have clearly proven that artichoke receptacle provided the highest therapeutic potential against liver and kidney damages. In fact, the said activities of stems and bracts were found higher than leaves, which is the officially recognized herbal drug. Since artichoke receptacles have economical value as a healthy vegetable in gastronomy, stems and bracts, as secondary products, should be preferred instead of leaves to prepare more efficient extracts to support liver homeostasis.

Data are mean $\pm$ SEM. Statistical analysis: Two-way ANOVA followed by Tukey Multiple Comparison test (^a, ^b, ^c, ^d, ^e, ^f, and ^g $p < 0.05$ as ^ap vs. Res; ^bp vs. OB; ^cp vs. IB; ^dp vs. St; ^ep vs. Lf; ^fp vs. Cont.; ^gp vs. SM extracts.).

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